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IT4010 Research Project – 2026 Jul

Topic Assessment Form

V2.2

Project ID:

Group ID : [J26-DS-343]

1. Topic (12 words max)

EuqVid: A Gender Bias Detection Metric for Video Datasets Using Visual and Temporal Features

2. Research group the project belongs to

CoEAI - Centre of Excellence for AI

3. Specialization of the project belongs to

Data Science (DS)

4. If a continuation of a previous project:

Project ID	N/A
Year	N/A

5. Brief description of the research problem including references (300 words max) – references not included in word count.

Video-based AI systems are increasingly deployed in high-stakes domains including action recognition, healthcare diagnostics, and content moderation. These systems are trained on large-scale video datasets that frequently contain systematic gender bias in how male and female subjects are visually and temporally represented. Despite growing awareness of AI fairness, no pre-training, model-agnostic metric currently exists to quantify gender bias in video datasets before model development begins.

Researchers, dataset curators, and developers of video-based AI products (the affected persona) currently have no standardized method to verify whether a video dataset under-represents one gender through unequal screen time, face visibility, spatial framing, or speaking duration. This directly affects end-users of downstream systems — for example, individuals misclassified by gender-biased deepfake detectors or video-based diagnostic tools.

The impact is technical (biased datasets propagate unequal model accuracy across gender subgroups), social (unequal treatment of genders in automated decisions), and ethical (lack of transparency in AI fairness auditing). Trinh and Liu (2021) found FaceForensics++, a widely used deepfake-detection dataset, is overwhelmingly composed of Caucasian and predominantly female subjects, with downstream detectors showing up to 10.7% error-rate differences across subgroups. Nadimpalli and Rattani (2022) confirm that gender-imbalanced training sets measurably degrade detector fairness.

The Geena Davis Inclusion Quotient, applied across 200+ commercial films, found male subjects receive roughly double the screen time and speaking time of female subjects (Geena Davis Institute, 2016–2019).

This project falls under the CoEAI (Centre of Excellence for AI) research cluster, addressing the Responsible AI / Bias Detection industry vertical, and directly extends the gender-bias-detection methodology established in EuqAud (Jayawardena et al., IEEE Access, 2026) — supervised by Dr. Prasanna S. Haddela — from the audio modality into video. This aligns with SDG 5 (Gender Equality), SDG 10 (Reduced Inequalities), and SDG 16 (Peace, Justice, and Strong Institutions).

6. Brief description of Existing Research and Systems (300 words max)

Existing products/systems include the Geena Davis Inclusion Quotient (GD-IQ), which measures screen time and speaking time gender disparities in commercial films, but is a proprietary descriptive analytics tool — not a machine-learning dataset-auditing metric. The Gender Balanced DeepFake Dataset (GBDF, Nadimpalli & Rattani, 2022) annotates gender on FaceForensics++ and Celeb-DF to address sample-count imbalance, but does not produce a continuous, traceable bias score. FairFace, RFW, and BFW provide balanced face image datasets but are static releases, not auditing frameworks applicable to video. EuqAud (Jayawardena et al., IEEE Access, 2026) provides a validated polynomial Ridge regression-based gender bias score for audio datasets, explicitly identifying video as unaddressed future work.

Existing research includes Buolamwini and Gebru (2018), who demonstrated intersectional accuracy disparities in commercial gender classifiers. Trinh and Liu (2021) found FaceForensics++ to be overwhelmingly Caucasian and predominantly female, with downstream deepfake detectors showing up to 10.7% error-rate differences across subgroups. Suresh and Guttag (2021) formalized how dataset composition propagates model harm. Attanasio et al. (2024) showed multilingual video-audio models exhibit systematic gender performance gaps across languages.

Gaps identified: (1) GD-IQ is a descriptive analytics tool, not a regression-derived, model-agnostic dataset-auditing metric; (2) GBDF and similar tools rebalance sample counts but do not score within-video representational bias such as screen time and spatial prominence; (3) EuqAud is restricted to the audio modality and explicitly identifies video as unaddressed future work; (4) no existing framework unifies visual presence, visual prominence, and temporal dynamics into a single pre-training, traceable, closed-form regression-based bias score for video datasets.

This project falls under the CoEAI (Centre of Excellence for AI) research cluster, addressing the Responsible AI / Bias Detection industry vertical, and directly extends the gender-bias-detection methodology established in EuqAud (Jayawardena et al., IEEE Access, 2026) — supervised by Dr. Prasanna S. Haddela — from the audio modality into video.

7. Brief description of the solution's nature, including a conceptual diagram (maximum 500 words).

EuqVid is a pre-training, model-agnostic gender bias detection metric for video datasets, extending the polynomial Ridge regression methodology of EuqAud (Jayawardena et al., IEEE Access, 2026) from the audio modality into video. EuqVid addresses all four identified gaps: it produces a continuous, traceable bias score (unlike dataset-rebalancing tools), it is designed as an ML pipeline-compatible auditing metric (unlike descriptive analytics tools such as GD-IQ), it operates on video-specific visual and temporal features (unlike EuqAud which covers audio only), and it unifies three dimensions of representational bias — visual presence, visual prominence, and temporal dynamics — into a single closed-form regression equation.

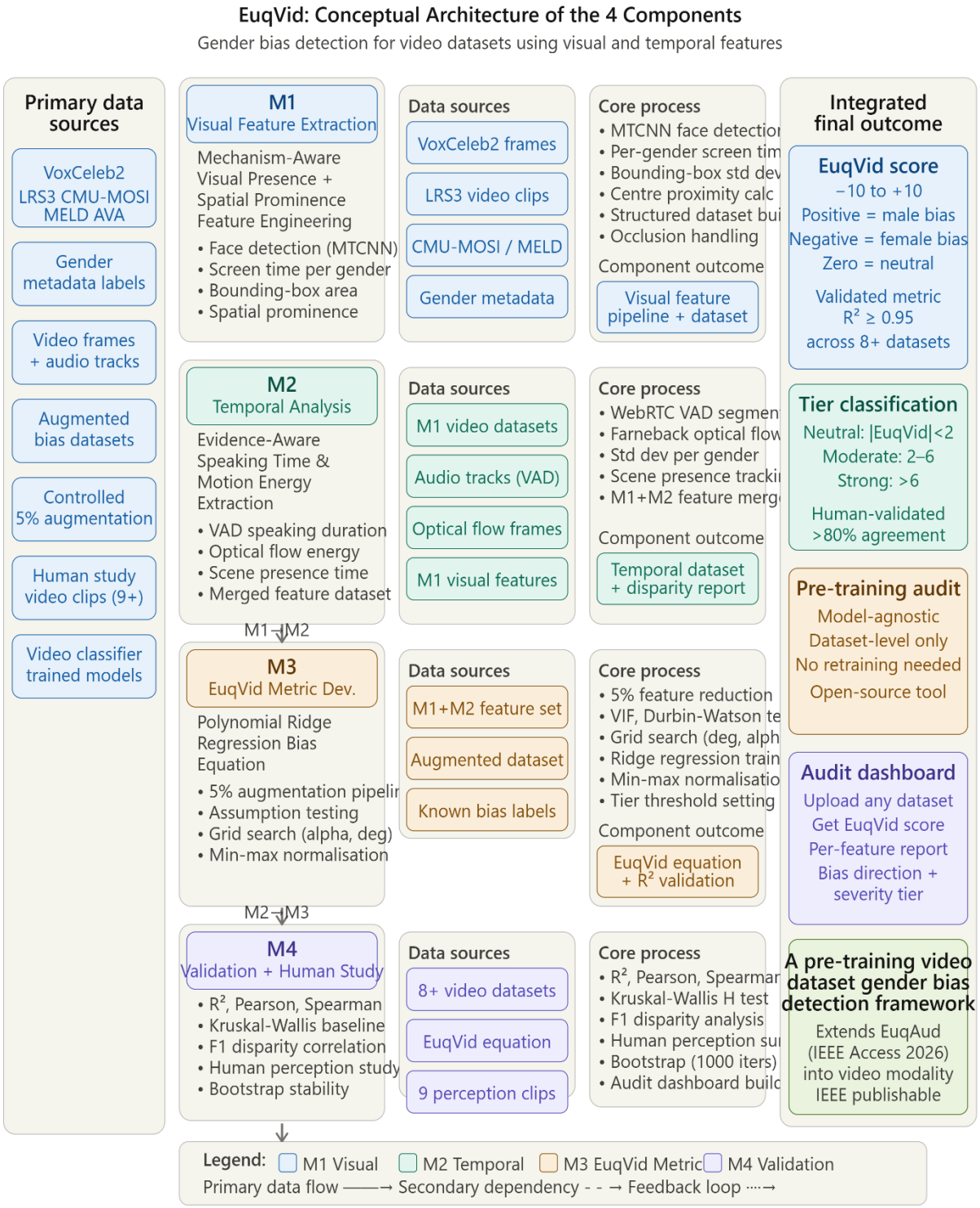
The solution comprises four linked components. First, M1 (Visual Feature Extraction and Bias Quantification) deploys MTCNN/RetinaFace for face detection and extracts per-gender screen time, bounding-box area, spatial framing, and standard deviations across 8+ public video datasets, building a structured visual feature dataset as the foundation for metric development. Second, M2 (Temporal Analysis — Speaking Time and Motion Energy Extraction) extracts per-gender speaking duration using lip-activity detection and WebRTC VAD, and per-gender motion energy via Farneback optical flow, normalising inherent gender differences before merging temporal features with M1 output into a unified dataset. Third, M3 (EuqVid Metric Development — Polynomial Ridge Regression) builds a controlled feature-level augmentation pipeline using 5% incremental reduction mirroring EuqAud, runs regression assumption checks, performs grid search over polynomial degree and Ridge alpha, derives the fixed EuqVid closed-form equation, applies min-max normalisation to [−10, +10], defines Neutral/Moderate/Strong bias tiers, and validates stability through 1,000-iteration bootstrap analysis. Fourth, M4 (Multi-Dataset Validation and Human Perception Study) validates EuqVid across 8+ public video datasets using R^2, Pearson, and Spearman correlation, compares EuqVid against baseline metrics using Kruskal-Wallis effect-size testing, correlates EuqVid tiers with downstream F1-score disparities, conducts a 9-clip human perception study, and builds a Streamlit audit dashboard.

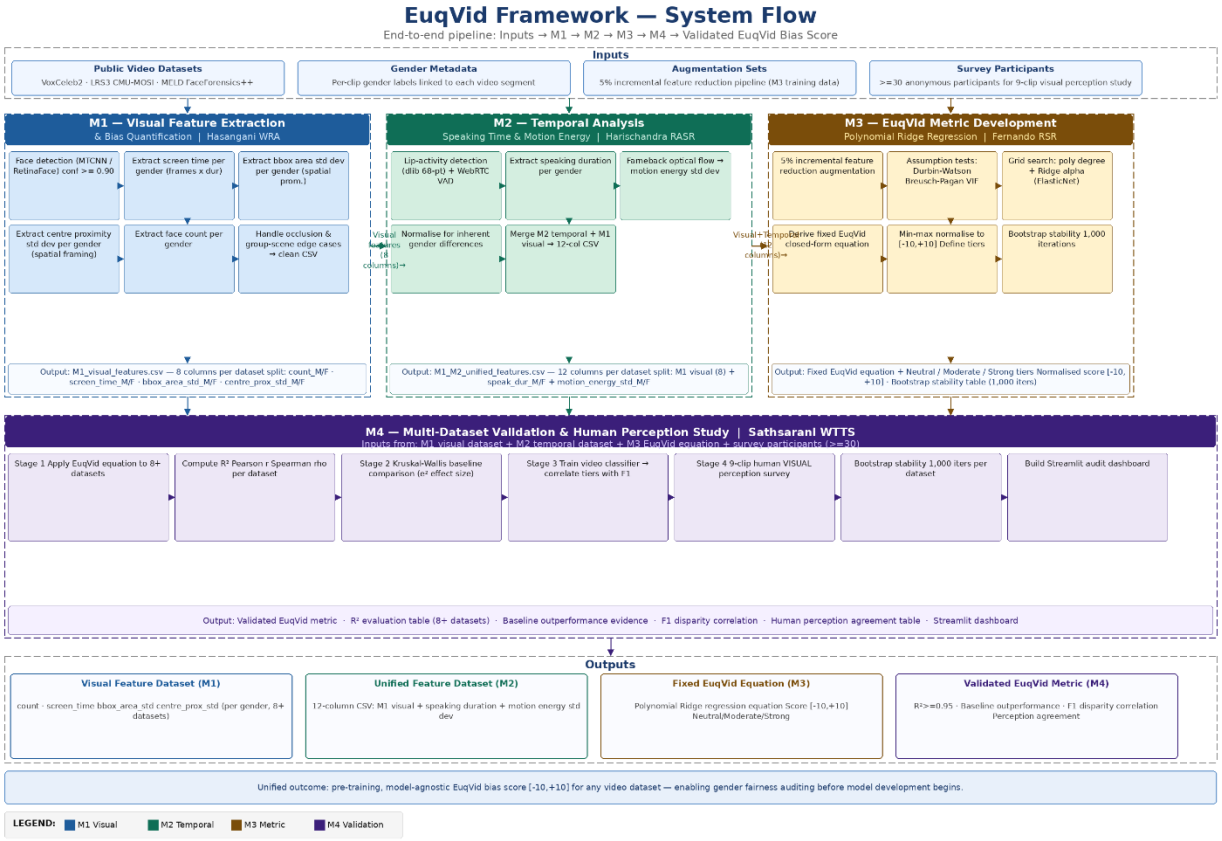
System Perspective

Inputs: Raw video frames and clips from public datasets (VoxCeleb2, LRS3, CMU-MOSI, MELD, FaceForensics++); per-speaker gender metadata; controlled augmentation feature sets generated at 5% incremental reduction steps; anonymous human participant responses for the 9-clip perception survey.

Processes: Face detection and visual feature extraction (M1); lip-activity and optical flow temporal feature extraction (M2); polynomial Ridge regression model training on augmented datasets, assumption checking, grid search, and equation derivation (M3); multi-dataset regression evaluation, baseline comparison, downstream F1 correlation, bootstrap stability analysis, and human perception survey (M4).

Outputs: Structured visual feature dataset (M1); unified visual and temporal feature dataset (M2); fixed EuqVid closed-form polynomial Ridge regression equation with Neutral/Moderate/Strong bias tiers normalised to [−10, +10] (M3); validated EuqVid metric with R^2 evaluation report, baseline outperformance evidence, downstream F1 disparity correlation, human perception agreement table, and Streamlit audit dashboard (M4).


EuqVid: Conceptual Architecture of the 4 Components
Gender bias detection for video datasets using visual and temporal features
Primary data sources
VoxCeleb2 LRS3 CMU-MOSI MELD AVA
Gender metadata labels
Video frames + audio tracks
Augmented bias datasets
Controlled 5% augmentation
Human study video clips (9+)
Video classifier trained models
M1 Visual Feature Extraction
Mechanism-Aware Visual Presence + Spatial Prominence Feature Engineering
• Face detection (MTCNN)
• Screen time per gender
• Bounding-box area
• Spatial prominence
Data sources
VoxCeleb2 frames
LRS3 video clips
CMU-MOSI / MELD
Gender metadata
Core process
• MTCNN face detectio
• Per-gender screen tim
• Bounding-box std dev
• Centre proximity calc
• Structured dataset bui
• Occlusion handling
Component outcome
Visual feature pipeline + dataset
M2 Temporal Analysis
Evidence-Aware Speaking Time & Motion Energy Extraction
• VAD speaking duration
• Optical flow energy
• Scene presence time
• Merged feature dataset
Data sources
M1 video datasets
Audio tracks (VAD)
Optical flow frames
M1 visual features
Core process
• WebRTC VAD segmen
• Farneback optical flow
• Std dev per gender
• Scene presence tracki
• M1+M2 feature merg
Component outcome
Temporal dataset + disparity report
M1 → M2
M3 EuqVid Metric Dev.
Polynomial Ridge Regression Bias Equation
• 5% augmentation pipelir
• Assumption testing
• Grid search (alpha, deg)
• Min-max normalisation
Data sources
M1+M2 feature set
Augmented dataset
Known bias labels
Core process
• 5% feature reduction
• VIF, Durbin-Watson te
• Grid search (deg, alph
• Ridge regression train
• Min-max normalisatio
• Tier threshold setting
Component outcome
EuqVid equation + R² validation
M2 → M3
M4 Validation + Human Study
• R², Pearson, Spearman
• Kruskal-Wallis baseline
• F1 disparity correlation
• Human perception study
• Bootstrap stability
Data sources
8+ video datasets
EuqVid equation
9 perception clips
Core process
• R², Pearson, Spearmar
• Kruskal-Wallis H test
• F1 disparity analysis
• Human perception su
• Bootstrap (1000 iters)
• Audit dashboard build
Integrated final outcome
EuqVid score
−10 to +10
Positive = male bias
Negative = female bias
Zero = neutral
Validated metric R² ≥ 0.95 across 8+ datasets
Tier classification
Neutral: |EuqVid|<2
Moderate: 2–6
Strong: >6
Human-validated >80% agreement
Pre-training audit
Model-agnostic
Dataset-level only
No retraining needed
Open-source tool
Audit dashboard
Upload any dataset
Get EuqVid score
Per-feature report
Bias direction + severity tier
A pre-training video dataset gender bias detection framework
Extends EuqAud (IEEE Access 2026) into video modality
IEEE publishable
Legend: M1 Visual M2 Temporal M3 EuqVid Metric M4 Validation
Primary data flow ——→ Secondary dependency - - → Feedback loop ····→


EuqVid Framework — System Flow
End-to-end pipeline: Inputs → M1 → M2 → M3 → M4 → Validated EuqVid Bias Score
Inputs
Public Video Datasets
VoxCeleb2 · LRS3 CMU-MOSI · MELD FaceForensics++
Gender Metadata
Per-clip gender labels linked to each video segment
Augmentation Sets
5% incremental feature reduction pipeline (M3 training data)
Survey Participants
>=30 anonymous participants for 9-clip visual perception study
M1 — Visual Feature Extraction
& Bias Quantification | Hasangani WRA
Face detection (MTCNN / RetinaFace) conf >= 0.90
Extract screen time per gender (frames x dur)
Extract bbox area std dev per gender (spatial prom.)
Extract centre proximity std dev per gender (spatial framing)
Extract face count per gender
Handle occlusion & group-scene edge cases → clean CSV
Output: M1_visual_features.csv — 8 columns per dataset split: count_M/F · screen_time_M/F · bbox_area_std_M/F · centre_prox_std_M/F
M2 — Temporal Analysis
Speaking Time & Motion Energy | Harischandra RASR
Lip-activity detection (dlib 68-pt) + WebRTC VAD
Extract speaking duration per gender
Farneback optical flow → motion energy std dev
Normalise for inherent gender differences
Merge M2 temporal + M1 visual → 12-col CSV
Output: M1_M2_unified_features.csv — 12 columns per dataset split: M1 visual (8) + speak_dur_M/F + motion_energy_std_M/F
M3 — EuqVid Metric Development
Polynomial Ridge Regression | Fernando RSR
5% incremental feature reduction augmentation
Assumption tests: Durbin-Watson Breusch-Pagan VIF
Grid search: poly degree + Ridge alpha (ElasticNet)
Derive fixed EuqVid closed-form equation
Min-max normalise to [-10,+10] Define tiers
Bootstrap stability 1,000 iterations
Output: Fixed EuqVid equation + Neutral / Moderate / Strong tiers Normalised score [-10, +10] · Bootstrap stability table (1,000 iters)
M4 — Multi-Dataset Validation & Human Perception Study | Sathsarani WTTS
Inputs from: M1 visual dataset + M2 temporal dataset + M3 EuqVid equation + survey participants (>=30)
Stage 1 Apply EuqVid equation to 8+ datasets
Compute R² Pearson r Spearman rho per dataset
Stage 2 Kruskal-Wallis baseline comparison (ε² effect size)
Stage 3 Train video classifier → correlate tiers with F1
Stage 4 9-clip human VISUAL perception survey
Bootstrap stability 1,000 iters per dataset
Build Streamlit audit dashboard
Output: Validated EuqVid metric · R² evaluation table (8+ datasets) · Baseline outperformance evidence · F1 disparity correlation · Human perception agreement table · Streamlit dashboard
Outputs
Visual Feature Dataset (M1)
count · screen_time bbox_area_std centre_prox_std (per gender, 8+ datasets)
Unified Feature Dataset (M2)
12-column CSV: M1 visual + speaking duration + motion energy std dev
Fixed EuqVid Equation (M3)
Polynomial Ridge regression equation Score [-10,+10] Neutral/Moderate/Strong
Validated EuqVid Metric (M4)
R²>=0.95 · Baseline outperformance · F1 disparity correlation Perception agreement
Unified outcome: pre-training, model-agnostic EuqVid bias score [-10,+10] for any video dataset — enabling gender fairness auditing before model development begins.
LEGEND: M1 Visual M2 Temporal M3 Metric M4 Validation

8. Brief overview of the availability of necessary specialized domain expertise, knowledge, and data. (500 words max)

Domain Expertise

The project directly extends EuqAud (Jayawardena et al., IEEE Access, 2026), supervised by Dr. Prasanna S. Haddela, who serves as the project supervisor and provides direct domain expertise in responsible AI, bias detection methodology, and polynomial regression-based fairness metric development. This is a significant advantage — the team has access to the original paper's methodology, validation framework, and author-level guidance throughout the project. Each member additionally builds domain knowledge through targeted literature: Hasangani WRA (M1) studies computer vision fairness auditing through Gender Shades (Buolamwini & Gebru, 2018), GBDF (Nadimpalli & Rattani, 2022), and MTCNN/RetinaFace detection literature. Harischandra RASR (M2) gains expertise in temporal gender analysis through VoxCeleb research and GD-IQ media-analytics literature. Fernando RSR (M3) deepens expertise in polynomial regression and fairness metric derivation through direct study of the EuqAud methodology and regression assumption testing literature. Sathsarani WTTS (M4) builds expertise in multi-dataset validation design and human perception study methodology through EuqAud's four-stage validation framework.

Technical Skills

M1 — Python, OpenCV, MTCNN, RetinaFace, DeepFace for face detection and visual feature extraction. M2 — Farneback optical flow (OpenCV), WebRTC VAD, dlib for lip-activity detection and temporal feature extraction. M3 — scikit-learn (PolynomialFeatures, Ridge, GridSearchCV), statsmodels for regression assumption testing (Durbin-Watson, Breusch-Pagan, VIF), and SciPy for bootstrap stability analysis. M4 — SciPy statistical testing (Kruskal-Wallis, Pearson, Spearman), Google Forms for perception survey design, Streamlit for audit dashboard development, and SHAP for explainability. All members strengthen foundational ML and Python skills through the university curriculum and self-directed study via the Coursera Machine Learning Specialization and fast.ai practical deep learning courses.

Data Collection

All datasets required for EuqVid are publicly available and freely accessible. The primary datasets are VoxCeleb2, LRS3, CMU-MOSI, MELD, and FaceForensics++, all of which include gender metadata linking video clips to speaker gender. No primary data collection from human subjects is required for the core metric development. EuqVid's training data is generated synthetically through controlled feature-level augmentation of these public corpora, mirroring EuqAud's methodology exactly — preserving sample counts while systematically reducing one gender's feature values in 5% increments to create known ground-truth bias levels. The human perception study (M4) uses 9 short anonymised video clips sampled from the public datasets and administered through an anonymous voluntary online survey, requiring no collection of sensitive personal data.

Ethical Clearance

Formal ethical clearance is not required for this project. EuqVid uses only publicly available video datasets processed through automated pipelines with no collection of personal, sensitive, or identifiable data. The human perception survey is voluntary, fully anonymous, and collects no demographic or personal information beyond perception ratings. All dataset licences, access dates, and software versions will be documented for full reproducibility in accordance with responsible AI research practice.

9. Objectives and Novelty

Main Objective To develop EuqVid, a pre-training, model-agnostic gender bias detection metric for video datasets that quantifies gender representational disparities through visual and temporal features using polynomial Ridge regression, validated across 8+ public video datasets.			
Member Name with Registration No	Sub Objective	Tasks	Novelty
Hasangani WRA (IT23224384)	M1 — Visual Feature Extraction and Bias Quantification	1. Review prior visual gender-classification literature to justify feature selection, excluding context-dependent attributes such as facial expression. 2. Deploy MTCNN/RetinaFace for face detection across 8+ video datasets with confidence thresholding (≥0.90). 3. Extract per-gender screen time, bounding-box area, and spatial framing per frame. 4. Compute per-gender standard deviations of visual features. 5. Build structured visual feature dataset compatible with M3 regression pipeline. 6. Handle occlusion and group-scene edge cases.	First structured extraction of visual presence (screen time, face count), spatial prominence (bounding-box area), and spatial framing (centre proximity) — computed as standard deviations to isolate representational bias from natural gender appearance differences, mirroring EuqAud's feature-selection rationale — into a single regression-compatible video bias pipeline. No existing tool, including GD-IQ, GBDF, or EuqAud, extracts all three visual dimensions together for video dataset bias scoring.

		7. Validate extracted features through distribution plots per gender prior to handoff to M3.	
Harischandra RASR (IT23194908)	M2 — Temporal Analysis: Speaking Time and Motion Energy Extraction	1. Justify temporal feature selection with reference to GD-IQ speaking-time disparity findings and motion energy as a novel representational signal. 2. Extract per-gender speaking duration using dual-signal lip-activity detection (dlib) and WebRTC VAD. 3. Extract per-gender motion energy via Farneback optical flow standard deviation. 4. Compute standard deviations to normalise inherent gender differences. 5. Merge temporal features with M1 visual features into unified dataset. 6. Validate temporal pipeline across VoxCeleb2, MELD, and CMU-MOSI. 7. Confirm pipeline generalisability by testing across three distinct video styles (interview, TV dialogue, monologue).	First integration of per-gender optical flow motion energy AND dual-signal (lip-activity + audio VAD) speaking duration into a regression-based video dataset bias scoring pipeline, validated for generalisability across distinct video content styles. Temporal gender disparities documented by media-analytics literature have never been computationally fused into a machine-learning dataset audit metric.
Fernando RSR (IT23449282)	M3 — EuqVid Metric Development: Polynomial Ridge Regression Equation	1. Build controlled feature-level augmentation pipeline using 5% incremental reduction mirroring EuqAud. 2. Run regression assumption checks (Durbin-Watson, Breusch-Pagan, VIF). 3. Compute Pearson correlation coefficients between each feature and	First closed-form polynomial Ridge regression equation for video gender bias scoring derived from controlled visual and temporal feature augmentation, supported by Pearson correlation analysis identifying the strongest predictive features — directly extending EuqAud's audio

		the bias score to identify the strongest predictive features. 4. Perform grid search over polynomial degree and Ridge alpha. 5. Derive fixed EuqVid closed-form equation. 6. Apply min-max normalisation to [−10, +10] and define Neutral/Moderate/Strong tier thresholds. 7. Run bootstrap stability analysis (1,000 iterations).	methodology into the video modality, a gap EuqAud explicitly identified as future work in Section V of the paper.
Sathsarani WTTS (IT23204102)	M4 — Multi-Dataset Validation and Human Perception Study	1. Validate EuqVid across 8+ public video datasets using R^2, Pearson, and Spearman correlation. 2. Compare EuqVid against baseline metrics (screen-time ratio, face-count ratio) using Kruskal-Wallis effect-size testing. 3. Train video classifier and correlate EuqVid tiers with downstream F1-score disparities. 4. Recruit a minimum of 15–20 participants for statistically meaningful human perception results. 5. Design and conduct a 9-clip anonymous human perception study spanning all three bias tiers. 6. Perform bootstrap stability analysis. 7. Build Streamlit audit dashboard for researcher-facing dataset auditing.	First four-stage validation of a video dataset gender bias metric mirroring EuqAud's validation rigour: regression-level R^2 evaluation, baseline outperformance, downstream F1 disparity correlation, and a statistically powered human perception study with visual (not auditory) stimuli — no prior video fairness metric has been validated across all four stages simultaneously.

10. <u>This section is to be completed by the Supervisor and the Co-supervisor of the Project.</u>

a) This research topic possesses a comprehensive scope suitable for a final-year project.

Yes		No	

b) The proposed topic exhibits novelty.

Yes		No	

c) The student group can successfully execute the proposed project.

Yes		No	

d) The proposed sub-objectives reflect the students' areas of specialization.

Yes		No	

e) Data required for the project is available.

Yes		No	

f) Any other comments:

	Title	First Name	Last Name	Signature
Supervisor*				
Co-Supervisor*				
External Supervisor				
Summary of external supervisor's (if any) experience and expertise				

***-** At this stage, identifying the Co-Supervisor is compulsory. Students should reach out to a staff member, with guidance from their supervisor, to serve as the co-supervisor for this project.